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Paper Code : 21307

F–407

B. C. A. (Second Semester) EXAMINATION, 2018

(New Course)

Paper No. BCA—N–202

DISCRETE MATHEMATICS

Time : Three Hours] [*Maximum Marks : 70*

Note : Attempt any *five* questions.

1. (a) Show that :

$$p \to q, q \to r \vdash p \to r$$

is valid argument.

(b) Explain the concept of tautology and contradiction using suitable example.

2. (a) How is predicate logic different from propositional logic ?

(b) Draw truth table for :

(i) $P \uparrow Q \uparrow R$

(ii) $\neg (p \wedge \neg q) \to (p \vee \neg r)$

3. (a) What are the logic gates ? Derive AND and OR gate using them/universal gates.

(b) Draw logic/ckt diagram to represent the Boolean expression :

(i) $(x_1 x_2)' + (x_1' x_2')$

(ii) $(x_1 + x_2 + x_3) x_4 + x_5$

(iii) $(x_1')'$

4. (a) What are the normal forms ? Convert the following expression into DNF and CNF :

$$(p \to q) \wedge (\neg p \vee \neg q)$$

(b) What is K-map ? Draw a structure of 4-variable K-map.

5. (a) Write and explain the Dijkstra's algorithm for shortest path problem.

(b) Define the following terms :

(i) Tree and Rooted tree

(ii) Bipartite graph

(iii) Complete graph

(iv) Regular graph

(v) Null graph

(vi) Distance in a graph

(vii) Path in a graph

6. (a) Let f: be a function defined as $f(x) = 2x + 1$ and g : be a function defined as $g(x) = (x+1)^2$. Determine the following :

(i) $f \circ g$

(ii) $g \circ f$

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(b) What are recursively defined functions ? Give suitable example.

7. (a) Let A = {5, 6, 7, 8}. Give example of relations which is :

(i) Reflexive

(ii) Symmetric

(iii) Transitive

(iv) Antisymmetric

(b) What do you mean by term "POSET" ? Describe an equivalence relation.

8. Write short notes on any *four* of the following :

(i) Planer graph

(ii) Graph colouring

(iii) Conditional statements

(iv) Laws of set theory

(v) Injective and bijective function

(vi) Digraph

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